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None disclosed.

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Racial disparities in diagnosis of invasive melanoma among veterans: A comparative analysis with the Surveillance, Epidemiology, and End Results Program

To the Editor: Melanoma was the fourth most common cancer in male Veterans Affairs (VA) patients in 2010.[1] Prior surveys found that 11% to 13% of US active-duty personnel routinely use sunscreen despite significant occupational sun exposure.[2] Racial disparities are important concerns in the VA and elsewhere. In this study, we compare stage distribution of melanoma at presentation, a critical prognostic factor, among white and nonwhite patients in the VA and in the general US population.

Invasive cutaneous melanoma cases from 2000 to 2019 were identified in the VA corporate data warehouse and the Surveillance, Epidemiology and End Results Program (SEER), which collects cancer incidence and survival data. Cases with an age <20 (due to absence of veterans <20), unknown histology, and melanoma in situ were excluded. Due to the small proportion of women in the at-risk veteran population, we restricted our analyses to men. Self-reported race information was collected for all cases. Two-tailed z tests were performed to test the difference in proportions of melanoma stages between the veteran population and the general population.

We identified 44,077 cases of invasive melanoma in the VA and 217,030 in SEER. Racial disparities in melanoma staging were substantially less pronounced in the VA than in SEER. In the VA, localized disease represented 77.9% of melanomas among whites, versus 71.0% among nonwhites. In SEER, the corresponding numbers were 80.7% versus 61.7%, over double the VA disparity ($P < .0001$). Similarly, the disparity between whites and nonwhites observed for metastatic disease at presentation (regional or distant) in the VA was lower than the disparity observed in SEER ($P < .0001$). Distant metastatic disease at presentation represented 6.1% of melanomas among whites versus 8.6% among nonwhites in the VA, while in SEER it represented 4.8% of melanomas among whites versus 11.3% of melanomas in nonwhites, more than double the VA disparity ($P < .0001$) (Table I).

The trend of a lower racial disparity in the VA in the proportion of melanomas with local disease and in the proportion of distant metastasis at presentation was observed across age groups (Table II). These differences between the VA and SEER were less marked in those >65 in age. Notably, the differences between VA and SEER in racial disparities among those >65 in age were still significant for localized disease ($P < .0001$) and for distant metastasis ($P < .0001$). Nevertheless, in the VA, the racial disparities were less in those <65 than those who were older ($P < .0001$ for localized disease; $P < .0001$ for distant metastasis).

Strengths of this study include its large population size. Limitations of this study include the predominantly elderly and male VA population, potentially underreported utilization of non-VA dermatologic care and variation in geographic regions covered by each database. The current results suggest that the

Table I. SEER summary stage of melanoma in males in VA and SEER cohorts by race

	VA cases (%)†			SEER cases (%)†			
Seer summary stage	Overall*	White	Nonwhite	Overall*	White	Nonwhite	P value‡
Localized only	28,023 (71.4)	27,020 (77.9)	1003 (71.0)	149,928 (80.5)	143,062 (80.7)	1211 (61.5)	<.0001
Regional	4039 (10.3)	3861 (11.1)	178 (12.6)	16,930 (9.1)	16,454 (9.3)	404 (20.5)	<.0001
Distant site(s)/node(s) involved	2251 (5.7)	2130 (6.1)	121 (8.6)	8837 (4.7)	8581 (4.8)	232 (11.8)	<.0001
Unknown	4928 (12.6)	1665 (4.8)	110 (7.8)	10,481 (5.6)	9200 (5.2)	121 (6.1)	<.0001
Total	39,241 (100)	34,676 (100)	1412 (100)	186,176 (100)	177,297 (100)	1968 (100)	

SEER, Surveillance, Epidemiology, and End Results; *VA*, Veterans Affairs.
*Overall numbers include cases with unknown or unavailable race information.
†In the VA, 88.4% of melanomas were observed in white patients and 3.6% were observed in nonwhite patients. In SEER, 95.2% of melanomas were observed in white patients and 1.1% were observed in nonwhite patients.
‡Compares racial disparities (measured by the absolute difference between white and nonwhite patients).

Table II. SEER summary stage of melanoma in males in Veterans Affairs and SEER cohorts by race and age

		VA CDW		SEER		
Age category	SEER stage	White	Nonwhite	White	Nonwhite	P value*
<50	Localized	2088 (69.2)	121 (64.7)	20944 (81.1)	231 (60.0)	<.0001
	Regional	373 (12.4)	30 (16.0)	2769 (10.7)	86 (22.3)	<.0001
	Distant	243 (8.1)	18 (9.6)	1108 (4.3)	48 (12.5)	<.0001
	Unknown	314 (10.4)	18 (9.6)	1009 (3.9)	20 (5.2)	.233
	Total	3018 (100)	187 (100)	25830 (100)	385 (100)	
50-64	Localized	8218 (76.3)	343 (73.6)	44036 (81.3)	378 (61.0)	<.0001
	Regional	1245 (11.6)	53 (11.4)	5029 (9.3)	128 (20.6)	<.0001
	Distant	762 (7.1)	36 (7.7)	2688 (5.0)	78 (12.6)	<.0001
	Unknown	545 (5.1)	34 (7.3)	2417 (4.5)	36 (5.8)	.053
	Total	10770 (100)	466 (100)	54170 (100)	620 (100)	
65-80	Localized	12037 (80.1)	392 (72.5)	53726 (81.1)	412 (61.3)	<.0001
	Regional	1595 (10.6)	69 (12.8)	5599 (8.5)	139 (20.7)	<.0001
	Distant	800 (5.3)	47 (8.7)	3253 (4.9)	78 (11.6)	<.0001
	Unknown	604 (4.0)	33 (6.1)	3679 (5.6)	43 (6.4)	.001
	Total	15036 (100)	541 (100)	66257 (100)	672 (100)	
>80	Localized	4677 (79.9)	147 (67.7)	24356 (78.5)	190 (65.3)	<.0001
	Regional	648 (11.1)	26 (12.0)	3057 (9.8)	51 (17.5)	<.0001
	Distant	325 (5.6)	19 (8.8)	1532 (4.9)	28 (9.6)	<.0001
	Unknown	202 (3.5)	25 (11.5)	2095 (6.7)	22 (7.6)	<.0001
	Total	5852 (100)	217 (100)	31040 (100)	291 (100)	

SEER, Surveillance, Epidemiology, and End Results; *VA CDW*, Veterans Affairs Corporate Data Warehouse.
*Compares racial disparities (measured by the absolute difference between white and nonwhite patients).

VA may be more effective in reducing racial disparities in melanoma stage at diagnosis, potentially due to all patients in the VA dataset having insured access to health care, regardless of socioeconomic status. Decreased racial disparities across systems in tumor staging observed in the >65 population may be related to the availability of Medicare to the older general population.

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Conflicts of interest

None disclosed.

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Trends in positive patch tests for formaldehyde-containing allergens found in personal care products

To the Editor: Allergic contact dermatitis due to personal care products (PCPs) is common. PCP ingredients are influenced by public opinion and manufacturing regulations.[1] In 2012, many American PCP manufacturers removed formaldehyde and formaldehyde releasers following the US Department of Health and Human Services designating formaldehyde as a carcinogen.[2] We hypothesized that PCP reformulations changed the prevalence of positive patch tests (PPTs) for allergens commonly found in PCPs.

The patch test results from a tertiary referral dermatitis clinic were prospectively collected. Patch test results for preservatives, fragrances, and acrylates that commonly cause allergic contact dermatitis from PCPs were analyzed. Patients with occupational exposure to biocides, such as painters, mechanics, and machinists, were excluded from the preservative analysis. Assemblers and beauty salon workers, including nail technicians, with possible occupational exposure to acrylates were excluded from acrylate analysis. Initial readings of patch tests occurred on day 4 and final readings on day 7. Patch test results from January 2002 to December 2012 were compared with those from January 2016 to December 2021. A 4-year washout period was used to allow time for product removal from retail circulation. PPTs between the 2 time periods were compared using the χ^2 analysis and Fisher's exact test.

In total, 1839 patients were patch tested to allergens commonly found in PCPs. PPTs for formaldehyde remained stable from 2002-2012 to 2016-2021 (7.49% vs 7.50%, respectively; $P = .994$). PPTs for all formaldehyde releasers decreased (3.19% vs 1.14%, respectively; $P < .001$) and were statistically significant for quaternium-15 (8.52% vs 3.20%, respectively; $P < .001$) and bronopol (2.50% vs 0.75%, respectively; $P = .03$). The prevalence of PPTs for toluenesulfonamide formaldehyde resin decreased but was not statistically significant (Table I).

All fragrances trended toward higher PPT prevalence in the 2016-2021 cohort, with a statistically significant increase in PPT for fragrance mix I (13.32% vs 26.92%, $P < .001$), propolis (4.24% vs 15.98%, $P < .001$), and cinnamic aldehyde (2.85% vs 5.91%, $P = .003$) (Table II). All acrylates tested trended toward higher PPT, with ethyl acrylate (0.80% vs 2.14%, $P = .0317$) and methyl methacrylate (0.59% vs 2.16%, $P = .008$) achieving statistical significance. The prevalence of PPTs for methylisothiazolinone (MI)/methylchloroisothiazolinone (MCI) increased (3.90% vs 11.95%, $P = .01$).

In those recently presenting to our dermatitis clinic, PPTs for formaldehyde releasers significantly decreased, whereas PPTs for fragrances, acrylates, and MI/MCI significantly increased. Our findings of increased PPTs for fragrances are in line with recent national patch testing trends.[3] Our persistent MI/MCI positivity is likely due to MI alone remaining in PCP in the United States despite restrictions in the European Union and Canada in 2013 and 2015.[4] PPTs for formaldehyde remained remarkably stable over time, likely because of non-PCP sensitizers.

This study is limited by the fact that patients referred to an academic clinic may not be representative of the general population.

Our study highlights how changes by manufacturers and policymakers are correlated with changes in patch test findings. The decrease in formaldehyde releaser positivity is associated temporally with regulatory changes that resulted in reduction of PCPs containing formaldehyde. Similarly, increase in the prevalence of PPT for acrylates was associated temporally with acrylates replacing formaldehyde resin in many nail polishes.

These findings support the recommendation that patch test series should be updated to include more current common allergens.[5]

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